Occupation, Smoking, and Alcohol in the Epidemiology of Bladder Cancer

ROSS C. BROWNSON, PHD, JIAN C. CHANG, PHD, AND JAMES R. DAVIS, MS

Abstract: We conducted a case-control study to evaluate the effects of occupation, smoking, and alcohol consumption on bladder cancer risk. A total of 823 male cases and 2,469 age-matched controls were identified through the Missouri Cancer Registry. Relative risk estimates of 2.0 or greater were observed for janitors and cleaners, mechanics, miners, and printers. Current cigarette smoking was associated with a two-fold excess risk of bladder cancer, whereas alcohol consumption showed no association with bladder cancer risk. (*Am J Public Health* 1987; 77:1298–1300.)

Introduction

Bladder cancer was first linked to an occupational exposure before 1900.[1] During the past 20 years, an increased risk of bladder cancer has been identified for a variety of occupations including chemical and rubber workers, leather workers, painters, dye workers, truck drivers, and garage and gas station workers.[2–7] It has been estimated that occupational exposures may account for 18 to 35 per cent of bladder cancer incidence among various male populations.[7,8]

Numerous studies have found an approximate doubling in bladder cancer risk for cigarette smokers.[6,9–13] Although this increase in risk is relatively smaller than those for some other smoking-related tumors, the smoking and bladder cancer association is considered a causal one, especially among males.[14] Studies have generally failed to identify a relation between alcohol consumption and bladder cancer risk.[15–17] However, two recent reports have suggested bladder cancer associations for beer and high-proof spirits consumption[18] and for drinking among smokers.[19] Mommsen, *et al*,[20] found a relative risk estimate of 1.6 for alcohol use.

The present study was designed to evaluate further the effects of occupation, smoking, and alcohol consumption on bladder cancer risk.

Methods

Subjects were identified through the Missouri Cancer Registry maintained by the Missouri Department of Health. The registry has been collecting data on cancer cases from public and private hospitals since 1972. Reporting has been mandated by law since 1984. Cases were White male Missouri residents who were diagnosed with histologically confirmed bladder cancer (International Classification of Diseases for Oncology[21] [ICD-O] code 188) between July 1984 and June 1986. Cases were limited to White males owing to the small number of other racial/ethnic groups and the lack of occupational diversity among females. Controls were selected from other patients in the registry, after exclusion of smoking-related sites. For each case, three White male controls were frequency-matched by age group. A total of 823 eligible cases and 2,469 controls were identified. Table 1 presents the distribution of cancer sites among controls. Cancers of the genitourinary organs constituted the largest proportion of the control group, with prostate cancer as the single predominant type (40 per cent of controls).

Address reprint requests to Ross C. Brownson, PhD, Bureau of Cancer Epidemiology and Control, Missouri Department of Health, 201 Business Loop 70 West, P.O. Box 1268, Columbia, MO 65205. Mr. Davis is also affiliated with that Bureau; Dr. Chang is with the Missouri Cancer Registry, also in Columbia. This paper, submitted to the Journal March 24, 1987, was revised and accepted for publication May 27, 1987.

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The registry systematically collects data on smoking history, alcohol consumption, and occupational history. Reporting of these variables is mandatory for Missouri hospitals. Hospital registrars record information for cancer patients using a standardized protocol. Information from the hospitals is then coded and checked by the registry staff. Smoking and alcohol information are characterized according to status (i.e., never, former, or current use) and by average daily amount smoked or consumed. Longest held job is recorded as usual occupation and industry and coded with the 1980 US census codes.[22]

We used unconditional logistic regression models to obtain maximum likelihood estimates of the relative risk (RR) and 95 per cent confidence intervals (CI).[23,24] All results presented were based on logistic regression since it allowed for simultaneous adjustment for multiple risk factors.

For work history analyses, occupations and industries with six or more cases were compared with low-risk employment categories, similar to the approach used by Flanders and Rothman.[25] Low-risk categories were those in which workers were likely to have few occupational exposures to carcinogens. The low-risk occupations included professionals, managers, sales, and clerical workers (census codes 1–389). The low-risk industries were wholesale and retail trade, finance, insurance, real estate, business services, and professional services (census codes 500–742 and 812–892). Since prostate cancer and bladder cancer may share some occupational risks[26,27] and prostate cancer was the most common site in the control group, analyses were conducted separately for all controls (group 1) and for all controls except prostate cancer (group 2). For smoking and alcohol analyses, the lowest level of exposure was used as the reference level. Subjects with unknown or missing information were excluded from the relevant analyses.

Results

Cases and controls had similar age distributions. Over 90 per cent of the subjects were alive when information was obtained for the Missouri Cancer Registry. The results of logistic regression analyses by smoking history and alcohol

TABLE 1—Distribution of Controls by Cancer Site

Site (ICD-O code)	Number (%)
Digestive organs and peritoneum (150–159)	827 (33.5)
Hematopoietic and reticuloendothelial systems (169)	157 (6.4)
Skin (173)	243 (9.8)
Genitourinary organs (179–189)	1033 (41.8)
Nervous system (191–192)	71 (2.9)
Endocrine glands (193–194)	31 (1.3)
Lymph nodes (196)	107 (4.3)
Total	2469 (100.0)

TABLE 2—Adjusted Relative Risk Estimates* and 95% Confidence Intervals (CI) According to Smoking and Alcohol Consumption

Factor	No. of Cases	No. of Controls	RR (95% CI)
Smoking History			
Never smoked	247	945	1.0
Ex-smoker	148	443	1.2 (1.0–1.6)
Current smoker	299	641	1.9 (1.5–2.3)
<1 pack/day	132	314	1.6 (1.2–2.0)
1+ packs/day	167	327	2.1 (1.6–2.6)
Unknown	129	440	
Alcohol Use			
Never drank	346	1141	1.0
Ex-drinker	22	69	0.9 (0.5–1.5)
Current drinker	239	663	1.1 (0.9–1.3)
<2 drinks/day	176	450	1.2 (0.9–1.5)
2+ drinks/day	63	213	0.8 (0.6–1.1)
Unknown	216	596	

*Adjusted for age and smoking or alcohol use, when appropriate.

use are shown in Table 2. The RR of bladder cancer for ever-smokers was 1.5 (95 per cent CI 1.3–1.9). Current smokers had a RR of 1.9, with an increasing trend by level of smoking. No elevated risk was observed for previous alcohol use and there was no trend in risk by amount consumed. A RR of 1.1 (95 per cent CI 0.9–1.3) was estimated for ever-drinkers. Since consumption of alcohol has been found to be positively associated with cancer of the colon and rectum,[28] additional analyses were conducted after exclusion of these sites. The RR for alcohol use was unchanged after this exclusion.

The findings according to usual occupation are summarized in Table 3. Results are shown for occupational categories that had risk estimates greater than 1.0. Four occupations had RRs of 2.0 or more including janitors and cleaners, mechanics, mining machine operators, and printing machine operators. The results were generally similar whether comparing cases to control group 1 or to control group 2. Workers employed in metal mining had the largest risk of bladder cancer (RR = 1.8) among five different categories of usual industry with RRs greater than 1.0 (Table 4).

Discussion

Our study emphasized the importance of smoking and occupation in the risk of bladder cancer. We found a two-fold risk among current smokers with an increasing trend in risk by amount smoked, consistent with other studies.[14] As in the majority of previous reports,[15–17,28] there was little or no association between overall alcohol use and bladder cancer.

The occupational findings were in agreement with previous studies for several employment categories. We found a RR of 3.5 for mechanics, which compares with a relative risk estimate of 1.33 reported by Smith, *et al,*[5] and of 1.6 estimated by Howe, *et al.*[6] Mechanics are exposed to a wide range of potential carcinogens including exhaust fumes, solvents, and fuel oils. A recent study found a RR of 2.7 among men who worked with oil or gasoline.[20] Our finding of a RR of 3.1 for printing machine operators is close to the previously reported relative risk estimates of 3.0 and 2.33.[3,18] Another study found a relative risk of 5.0 associated with jobs that involved high exposure to printing inks.[29] Our data also showed slightly elevated RRs among plumbers and machinists, findings also reported in other studies.[3,6,30,31]

A previously unreported increased risk was observed for janitors and cleaners (RR = 3.5). This occupational group is exposed to solvents and cleaning agents (e.g., aromatic hydrocarbons, naphtha-derivatives).[32] An association between truck driving and bladder cancer has been reported recently.[3,4] However, our data showed only a slightly increased risk among truck drivers.

We found an increased risk of bladder cancer among men employed in mining, specifically mining machine operators. Wynder, *et al,*[30] found a larger proportion of coal miners in the case group than among controls, although no relative risk estimate was reported; a German study[18] reported a relative risk of 1.85 for miners.

As in all epidemiologic studies, selection and information bias must be considered when interpreting these findings. The advantages and disadvantages of the use of cancer controls have been discussed in detail.[33] We were concerned that bladder cancer and prostate cancer may share some occupational risks and therefore analyzed the data separately after excluding prostate cancer from the control group. Most effect estimates showed little difference whether based on the first or second control group. Many occupational studies have compared each employment category with all other occupations. Such analyses may underestimate risk and may fail to identify some high-risk occupations.[5] To reduce this potential bias, we conducted occupational analyses by comparing each category with occupations and industries deemed to be low risk. Since the occupational information available in this study reflected only usual (longest held) employment,

TABLE 3—Adjusted Relative Risk Estimates* and 95% Confidence Intervals (CI) According to Usual Occupation

Occupation	No. of Cases	Control Group 1**		Control Group 2***	
		No.	RR (95% CI)	No.	RR (95% CI)
"Low Risk"	141	507	1.0	330	1.0
Janitors and Cleaners	13	14	3.5 (1.6–7.7)	9	3.7 (1.5–9.0)
Mechanics, Unspecified	9	9	3.5 (1.4–9.1)	5	4.2 (1.4–12.9)
Plumbers	8	22	1.2 (0.5–2.8)	10	1.7 (0.6–4.4)
Mining Machine Operators	9	9	2.9 (1.1–7.5)	5	3.2 (1.0–9.8)
Metal Machinists	10	19	1.7 (0.8–3.8)	15	1.4 (0.6–3.2)
Printing Machine Operators	7	8	3.1 (1.1–8.9)	6	2.3 (0.8–7.2)
Truck Drivers, Heavy	18	49	1.2 (0.7–2.1)	34	1.1 (0.6–2.0)
Laborers, Except Construction	15	47	1.2 (0.6–2.1)	29	1.3 (0.7–2.5)

*Adjusted for age, smoking, and alcohol use.
**All controls.
***All controls, excluding prostate cancer.

TABLE 4—Adjusted Relative Risk Estimates* and 95% Confidence Intervals (CI) According to Usual Industry

Industry	No. of Cases	Control Group 1** No.	Control Group 1** RR (95% CI)	Control Group 2*** No.	Control Group 2*** RR (95% CI)
"Low Risk"	126	387	1.0	254	1.0
Metal Mining	6	8	1.8 (0.6–5.3)	4	2.2 (0.6–8.2)
Beverage Manufacturing	9	17	1.5 (0.7–3.5)	11	1.4 (0.6–3.5)
Industrial Chemical Manufacturing	6	18	1.1 (0.4–2.8)	16	0.8 (0.3–2.1)
Railroads	18	45	1.3 (0.7–2.3)	21	1.6 (0.8–3.1)
Trucking Service	17	37	1.2 (0.7–2.2)	29	1.1 (0.6–2.2)

*Adjusted for age, smoking, and alcohol use.
**All controls.
***All controls, excluding prostate cancer.

it is possible that effects from short-term occupational pursuits were missed. If misclassification bias was present, it would tend to underestimate the true risk.[34,35] And finally, since information for this study was drawn from a number of hospitals, the possibility of differential reporting existed. However, there is no reason to believe that exposure data were selectively biased for cases or controls.

Coffee drinking and artificial sweetener use have been identified as risk factors in several studies, although effect estimates have been low and inconsistent.[36] The evaluation of coffee and artificial sweetener consumption has been hampered by the apparently small or nonexistent effect of these substances, their widespread use in the general population, and the strong correlation between coffee drinking and tobacco use. Our study did not address coffee and artificial sweetener use.

ACKNOWLEDGMENT

The authors are grateful for the assistance provided by the staff of the Missouri Cancer Registry, Missouri Department of Health.

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